Tamer Mahmoud

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PROFESSIONAL SUMMARY

Software engineer with 10 years of experience building production systems: cloud services, data pipelines, microservices and APIs, and databases, including system design and untangling legacy systems. Ran detection pipelines handling millions of security events per day at 99.9%+ availability, and reconciliation pipelines over billions of ad events per month. Building with LLMs through independent projects: retrieval (RAG), vision models, structured output, and AI-assisted triage. AWS Certified Solutions Architect - Associate, which backs up my work in cloud architecture. Core stack: TypeScript, Node.js, Python, React, PostgreSQL, AWS.

CERTIFICATIONS

AWS Certified Solutions Architect - Associate | ***Amazon Web Services*** Aug 2026

PROFESSIONAL EXPERIENCE

Independent Projects and AWS Certification Apr 2026 - Present

- Earned AWS Certified Solutions Architect - Associate (Aug 2026), covering high-availability design, security, cost optimization, and the AWS Well-Architected Framework.
- Built AI Security Triage, live on AWS: Claude rates CloudTrail events by severity with structured output and related activity, and analysts review each finding in a React dashboard. Event-driven pipeline on S3, SQS, and Lambda (TypeScript, Python), Aurora Serverless v2 through the RDS Data API, and Cognito sign-in, all in Terraform and deployed by GitHub Actions over OIDC. Evals on 40 hand-labeled cases: 80% exact, 95% within one level, about $3 per 1,000 alerts; 16/16 prompt injections held.

Senior Software Engineer | ***The Media Trust*** Sep 2025 - Mar 2026

- Built historical backfills, daily syncs, and a rolling 7-day reconciliation pipeline across Lambda, Kinesis, PostgreSQL, MongoDB, GAM, and Amazon Athena, processing billions of ad events monthly and eliminating recurring manual cleanup for downstream reporting.
- Redesigned CloudWatch-to-Slack alerting for sync and blocker workflows supporting a 24/7 malware operations desk processing 2,300+ active infections daily; added structured diagnostic context and automated daily status summaries that reduced triage response time from hours to minutes.
- Designed and extended backend APIs powering the partner dashboard, adding multi-select filtering and external data delivery on top of GAM order data for 600+ publisher clients including Microsoft and Fox News, cutting manual report requests from the ops team.
- Overhauled the Docker-based local dev environment and rewrote onboarding documentation for a ~30-person engineering org, cutting new engineer ramp time from ~4 weeks to under 2 weeks.

Software Engineer | ***Expel*** Dec 2022 - Jun 2025

- Designed and operated ingestion and detection pipelines processing millions of security events per day across 100+ vendor integrations, implementing normalization, enrichment, and correlation logic that doubled throughput and maintained 99.9%+ pipeline availability.
- Built the Node.js API, hunt query execution engine, and React investigation UI for the Emerging Threat feature, taking investigations from hunt queries through structured findings and analyst exports and replacing a manual process that took days to weeks across multiple staff.
- Led a solo backward-compatible schema migration across 140+ files, preserving legacy routes and dual naming patterns, shipped in 3 weeks with zero downtime.
- Refactored PostgreSQL and BigQuery query patterns and indexing strategies, cutting slow queries by 60% and improving search response times by 50% during peak ingestion.
- Resolved a ticket for missing data in an Elasticsearch-backed table by tracing it to events from a Go microservice that were never handled, a long-standing tech debt item. Added Go handlers for the missing events, with tests and test documents, and added a test confirming the data was visible and displayed correctly in the table.

Full Stack Developer | ***Sales Boomerang*** Apr 2022 - Oct 2022

- Built backend and frontend features in Angular, React, TypeScript, Node.js, and Sequelize for a borrower intelligence platform serving 170+ mortgage lenders, supporting near-real-time alert generation across millions of borrower records.

- Built event-driven pipeline infrastructure using AWS CloudFormation, CloudWatch, Kinesis, SQS, and SNS, supporting automated borrower alerts during a period of 130%+ quarter-over-quarter alert volume growth.
- Automated repetitive development workflows and restructured shared code, reducing friction in the team's daily development cycle.

Web Consultant | ***Rural Sourcing, Inc. (client engagement)*** Aug 2021 - Apr 2022

- Built Python scripts to automate recurring report generation across multiple databases, eliminating manual data retrieval for the client team.
- Supported backend development for a client-facing portal, working with database-heavy stored procedures to maintain data integrity and keep production workflows running reliably.
- Containerized the team's local development environment with Docker, reducing setup variability across the team.

Software Engineer | ***Gallaudet University*** Oct 2012 - Aug 2020

- As sole engineer, built and maintained the core platform for the ASLPI program, a nationally recognized ASL proficiency evaluation system supporting thousands of examinees annually across faculty tenure, graduation, and career placement decisions.
- Rebuilt the ASLPI data layer from the ground up on MSSQL, replacing a legacy system with a scalable database storing 30+ years of evaluation records, rater reliability analytics, and longitudinal language learning histories.
- Designed and built a custom CMS and online assessment tooling that automated evaluation scheduling and administrative workflows for 15 ASLPI staff, cutting processes that previously took weeks down to hours.
- Introduced Node.js and React as part of a platform modernization and implemented WCAG 2.1 AA accessibility (keyboard navigation, ARIA labeling, screen-reader support) across all web interfaces, serving about 2,500 students and staff at a federally chartered university.

PROJECTS

Plant Health Predictor | *Node.js, Express, Claude API (vision)* github.com/TmoodGitHub/plant-health-predictor

- Multi-image plant health analysis: takes 2 to 10 photos over time, identifies the plant, scores each image, detects trends, and predicts the next 7 to 14 days. Model output validated field by field before use, uploads checked by file signature, per-IP rate limiting, and PDF report export.

AI Portfolio Assistant (RAG) | *Node.js, Python, OpenAI embeddings* vite-react-portfolio-lime.vercel.app

- Retrieval-augmented chatbot that answers recruiter questions from resume, GitHub, and career history content: embeds each question, picks the closest knowledge-base sections, drops weak matches below a measured similarity cutoff, and answers only from the rest, with a guardrailed system prompt, per-IP rate limiting, and serverless deployment.

tmoodgit | *Node.js* github.com/TmoodGitHub/tmoodgit

- Git implemented from scratch: content-addressable object storage (blobs, trees, commits) with SHA-1 hashing, a staging index, branch refs, checkout, and a line-level LCS diff engine behind a full CLI.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go, PHP, Java

Frontend: React, Angular, Next.js, Ember.js, Tailwind CSS

Backend & Architecture: Node.js, REST APIs, GraphQL (independent projects), gRPC, JWT authentication, Sequelize ORM, API design, event-driven architecture, microservices, distributed systems

Databases & Search: PostgreSQL, MySQL, MSSQL, MongoDB, BigQuery, Amazon Athena, Elasticsearch

Cloud & Infrastructure: AWS (Lambda, S3, CloudFormation, IAM, SQS, SNS, Kinesis, CloudFront, API Gateway, Aurora Serverless, Cognito, Secrets Manager), GCP (Pub/Sub), Terraform, Docker, Serverless Framework

AI & LLM: Anthropic Claude API, OpenAI API, embeddings, retrieval-augmented generation (RAG), prompt design, Claude Code, Cursor

Observability & CI/CD: CloudWatch, Datadog, GCP Logging, GitHub Actions, CircleCI, Git

Testing: Jest, Vitest, pytest, Mocha, Chai, Cypress, React Testing Library, QUnit

EDUCATION

Bachelor of Science in Information Technologies | *Rochester Institute of Technology*